HealthTwin AI — System Architecture

Synthetic-data glucose forecasting prototype

1. DATA GENERATION

generate_data.py

Synthetic patient profiles and hourly sensor readings

Outputs: CSV files

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2. PREPROCESSING

train_model.py

Merge patient and sensor data; create lag features and 2-hour target

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3. MODEL TRAINING

Scikit-learn RandomForestRegressor

Fit training split

Save model and metrics

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4. EVALUATION

evaluate_unseen_patients.py

Temporal and held-out synthetic-patient tests

Compare baseline

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5. STREAMLIT APP

app.py

Choose synthetic patient

View history and prediction; simulate new reading

OUTPUTS: glucose_model.joblib | metrics.json | unseen_patient_metrics.json | Streamlit dashboard

Synthetic data only. Not clinically validated; not for diagnosis or treatment.